

Complex Bash

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This handout is intended as a resource for practicing unconventional and advanced problems in complex bashing. I have also noticed that several handouts have problems that are *bashable*, but no one in their right mind would ever bash them, so I have tried to include problems where I believe a bash is the most efficient approach.

1 Techniques

1.1 Warm up

Problem 1

Let ABC be a triangle with incenter I and circumcircle Ω . Let D, E, F be the midpoints of the arcs $\widehat{BC}, \widehat{CA}, \widehat{AB}$ of Ω not containing A, B, C respectively. Let D' be the point of Ω diametrically opposite to D . Show that I, D' and the midpoint M of EF lie on a line.

Proof. Notice $a = x^2, b = y^2, c = z^2, d = -yz, e = -zx, f = -xy, j = -(xy + yz + zx), d' = yz$, and $m = \frac{-xy-yz}{2}$. Then,

$$\frac{2yz + xy + xz}{\frac{2yz+xy+xz}{2}} = 2 \in \mathbb{R}$$

□

Problem 2

The foot of the altitude from vertex A of acute triangle ABC is T_A . The ray drawn from A through the circumcentre O intersects BC at R_A . Let the midpoint of AR_A be F_A . Define $T_B, R_B, F_B, T_C,$

R_C, F_C similarly. Prove that $T_A F_A, T_B F_B, T_C F_C$ are concurrent.

Proof. Let (ABC) be the unit circle. $T_A = \frac{1}{2} \left(a + b + c - \frac{bc}{a} \right)$.

$$R_A = BC \cap AA' = \frac{a^2(b+c)}{a^2+bc}.$$

$$F_A = \frac{a+R_A}{2} = \frac{1}{2} \left(\frac{a(a^2+bc)+a^2(b+c)}{a^2+bc} \right).$$

Since

$$\frac{n_9 - F_A}{n_9 - T_A} = \frac{a(b+c)}{a^2+bc} = \frac{\overline{a(b+c)}}{a^2+bc}$$

T_A, F_A, N_9 are collinear. By symmetry, $T_A F_A, T_B F_B, T_C F_C$ concur at N_9 . □

1.2 Homothety

Problem 3

In triangle ABC with $CA = CB$, point E lies on the circumcircle of ABC such that $\angle ECB = 90^\circ$. The line through E parallel to CB intersects CA in F and AB in G . Prove that the centre of the circumcircle of triangle EGB lies on the circumcircle of triangle ECF .

Proof. Let O and P be the circumcentres of $\triangle ABC$ and $\triangle EGB$ respectively, and let EF intersect (ABC) again at T .

Clearly, E and T are the antipodes of B and C respectively.

Notice that $\angle FEO = \angle TEB = \angle TCB = \angle ACT = \angle FCO$, so C, E, F, O are concyclic.

Let's employ complex numbers with respect to (ABC) , with $a = \frac{c^2}{b}$, $e = -b$, $t = -c$.

The intersection formula gives $g = \frac{b^2+bc+2c^2}{b-c}$. Shifting by b and taking a homothety with scale factor $\frac{1}{2}$ gives $b' = b$, $g' = \frac{b^2+c^2}{b-c}$, $e' = 0$.

Now, using the circumcircle formula and shifting back gives $p = \frac{b(b+c)}{b-c}$. It suffices to prove C, E, O, P are concyclic $\iff \left(\frac{c-p}{e-p} \right) \times \frac{e}{c} = \frac{b}{c} + \frac{c}{b} \in \mathbb{R}$ Which is evidently true. □

Problem 4

Let ABC be an acute triangle with incentre I and $AB \neq AC$. Let lines BI and CI intersect the circumcircle of ABC at $P \neq B$ and $Q \neq C$, respectively. Consider points R and S such that $AQRB$ and $ACSP$ are parallelograms (with $AQ \parallel RB, AB \parallel QR, AC \parallel SP$, and $AP \parallel CS$). Let T be the point of intersection of lines RB and SC . Prove that points R, S, T , and I are

concylic.

Proof. Notice that $\angle RTS = \angle QAP$, so it suffices to prove $\angle RIS = \angle QAP$.

Now, work on the complex plane with $a = x^2, b = y^2, c = z^2$. So, $p = -xz, q = -xy, j = -xy - yz - zx, r = y^2 - xy - x^2, s = z^2 - xz - x^2$.

$$\left(\frac{r-j}{s-j} \right) \left(\frac{a-p}{a-q} \right) \in \mathbb{R}$$

$$\iff \frac{(x+y)(y-x+z)}{(x+z)(y-x+z)} \times \frac{x+z}{x+y} = 1 \in \mathbb{R}$$

which is trivially true. □

Problem 5

In triangle ABC , consider points A_1, A_2 on line BC such that A_1, B, C, A_2 are in that order and $A_1B = AC$ and $CA_2 = AB$. Similarly consider points B_1, B_2 on line AC , and C_1, C_2 on line AB . Prove that the orthocentres of triangles $A_1B_1C_1$ and $A_2B_2C_2$ are equidistant from the circumcentre of ABC .

Proof. Let DEF be the intouch triangle of $\triangle ABC$ with centre I , and let H_1, H_2, H be the orthocentres of $\triangle A_1B_1C_1, \triangle A_2B_2C_2, \triangle DEF$ respectively. Let O be the circumcentre of ABC .

Notice that $A_1, A_2, B_1, B_2, C_1, C_2$ are concyclic with centre I , since $DI = EI = FI$ and $DA_1 = DA_2 = EB_1 = EB_2 = FC_1 = FC_2 = s$, where s is the semiperimeter of $\triangle ABC$

H is the midpoint of $\overline{H_1H_2}$

Proof. Let $A_1A_2B_1B_2C_1C_2$ be the unit circle with $A_1 = a, B_1 = b, C_1 = c, A_2 = \omega a, B_2 = \omega b, C_2 = \omega c$ since $A_1A_2 = B_1B_2 = C_1C_2$.

D, E, F are the midpoints of A_1A_2, B_1B_2, C_1C_2 respectively, so $d = \frac{a(1+\omega)}{2}$ and so on.

Taking a homothety with scale factor $\frac{2}{1+\omega}$ and scaling back gives $h = \frac{(1+\omega)(a+b+c)}{2}$, which is precisely the midpoint of $\overline{H_1H_2}$. □

Additionally, $HI \perp H_1H_2$ since $\frac{(1+\omega)(a+b+c)}{(1-\omega)(a+b+c)} = \frac{1+\omega}{2(1-\omega)}$ is purely imaginary.

To finish, note that the inversion with respect to the incircle takes O to the nine-point center of $\triangle DEF$, which lies on IH . $\square$

Problem 6

Let $\triangle ABC$ be an acute angled scalene triangle with circumcircle ω . Let O and H be the circumcenter and orthocenter of $\triangle ABC$, respectively. Let E, F and Q be points on segments AB, AC and ω , respectively, such that

$$\angle BHE = \angle CHF = \angle AQH = 90^\circ.$$

Prove that OQ and AH intersect on the circumcircle of $\triangle AEF$.

Proof. Let the antipode of Q be Q' , and let AH meet ω again at A' . Let T be $QQ' \cap AA'$. We need T to lie on (AEF) .

Let (ABC) be the unit circle, with $O = 0$ and $h = a + b + c$.

$$\frac{e - a}{e - b} = \frac{\bar{e}a - 1}{\bar{e}b - 1} \times \frac{b}{a} \implies \bar{e} = \frac{a + b - e}{ab}$$

$$\frac{h - b}{h - e} = \frac{a + c}{a + b + c - e} = -\frac{b(a + c)}{ab + ec} \implies e = \frac{b(2a + b + c)}{b - c}$$

Similarly $f = \frac{c(2a + b + c)}{c - b}$.

Now,

$$\frac{q - a}{q - a - b - c} = \frac{bc(q - a)}{abc - q(ab + bc + ca)} \implies q = \frac{2abc + b^2c + bc^2}{ab + 2bc + ac}$$

$$a' = \frac{a - a - b - c}{\frac{ab + bc + ca}{bc} - 1} = -\frac{bc}{a}$$

To compute T , notice

$$t = \frac{qq'(a + a') - aa'(q + q')}{qq' - aa'} = \frac{q^2(a^2 - bc)}{a(q^2 - bc)} = -\frac{bc(2a + b + c)^2}{a(b - c)^2}$$

Take a homothety with scale factor $\frac{b-c}{2a+b+c}$ to get $e = b$, $f = -c$, $a = \frac{a(b-c)}{2a+b+c}$, $t = -\frac{bc(2a+b+c)}{a(b-c)}$.

$$\frac{a - f}{e - f} = \frac{a + c}{2a + b + c}$$

and

$$\frac{a-t}{e-t} = \frac{a^2(b+c) + 4abc + bc(b+c)}{b(2a+b+c)(a+c)}$$

So,

$$\frac{a-f}{e-f} \times \frac{e-t}{a-t} = \frac{b(a+c)^2}{a^2(b+c) + 4abc + bc(b+c)}$$

which is equal to its conjugate, so we are done. $\square$

1.3 Inversion

Problem 7

Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA , and AB at D, E , and F , respectively. The line through D perpendicular to EF meets ω at R . Line AR meets ω again at P . The circumcircles of triangle PCE and PBF meet again at Q .

Prove that lines DI and PQ meet on the line through A perpendicular to AI .

Proof. Let D' be the antipode of D on ω , and let T be a point on DI such that $AT \parallel EF$. It suffices to prove P, Q, T are collinear.

Invert with respect to the incircle. We now have the following problem, removing $*$ s:

Let triangle D, E, F have midpoints A, B, C . Define P, Q, R, D' similar to the original problem, and let $Q = (BFP) \cap (CEP)$. Prove that Q lies on the nine-point circle Ω of ADD' .

P, A, D' collinear

Proof. Work on the complex plane. Let $D = d, E = e$ and $F = f$ so $r = \frac{-ef}{d}$.

Now, we compute

$$p = \frac{r-a}{r\bar{a}-1} = \frac{2def + ef(e+f)}{d(e+f) + 2ef}$$

So, we obtain

$$\frac{1}{2} \times \frac{d' - p}{d' - a} = \frac{(d^2 + ef)(e+f) + 4def}{(2d + e + f)(2ef + de + df)}$$

which is equal to its conjugate. $\square$

Additionally, $Q \in BC$ since

$$\angle PQB = \angle PFD = \angle PEC = \angle PQC$$

To finish, notice that the midpoint M of AD and BC lies on Ω , and

$$\angle MIP = \angle D'PI = \angle ID'P = \angle DEP = \angle MQP$$

□

Problem 8

Let ABC be an acute-angled triangle with circumcircle ω and circumcentre O . Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X , and BO meet AC at Y .

Prove that the circumcircles of triangles BXD and CYE have an intersection lie on line AO .

Proof. We claim the intersection is $P = AO \cap (BOC)$. For this, we use complex numbers with unit circle (ABC) .

Notice that inversion at O sends P to $AO \cap BC$, which is just

$$\frac{bc(a-a) + a^2(b+c)}{bc + a^2} = \frac{a^2(b+c)}{a^2 + bc}$$

by the intersection formula.

Inverting back gives us

$$p = \left(\frac{a^2(b+c)}{a^2 + bc} \right)^{-1} = \frac{a^2 + bc}{b+c}$$

Additionally,

$$\frac{b-d}{a-c} = \frac{\frac{1}{b} - \frac{1}{d}}{\frac{1}{a} - \frac{1}{c}} \implies d = -\frac{ac}{b}$$

and

$$x = \frac{ab(c-c) - (-c^2)(a+b)}{ab - (-c^2)} = \frac{c^2(a+b)}{ab + c^2}$$

Finally,

$$\frac{(b-x)(d-t)}{(d-x)(b-t)} = \frac{a(b^2 - c^2)}{c(b^2 - a^2)}$$

Which is equal to its conjugate. So, we are done by symmetry. $\square$

2 Practise Problems

I am going to be omitting the easy section, since this is *not* meant to be a first course in complex bashing.

2.1 Medium

- 1 Let ABC be a triangle with $AB < AC < BC$. Let the incenter and incircle of triangle ABC be I and ω , respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω . Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω . Let AI intersect the circumcircle of triangle ABC at $P \neq A$. Let K and L be the midpoints of AC and AB , respectively. Prove that $\angle KIL + \angle YPX = 180^\circ$.
- 2 Let ABC be a triangle and D, E, F be the midpoints of arcs BC, CA, AB on the circumcircle. Line ℓ_a passes through the feet of the perpendiculars from A to DB and DC . Line m_a passes through the feet of the perpendiculars from D to AB and AC . Let A_1 denote the intersection of lines ℓ_a and m_a . Define points B_1 and C_1 similarly. Prove that triangle DEF and $A_1B_1C_1$ are similar to each other.
- 3 Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.
- 4 Let ABC be a non-isosceles triangle with incenter I whose incircle is tangent to $\overline{BC}, \overline{CA}, \overline{AB}$ at D, E, F , respectively. Denote by M the midpoint of $\overline{BC}$. Let Q be a point on the incircle such that $\angle AQD = 90^\circ$. Let P be the point inside the triangle on line AI for which $MD = MP$. Prove that either $\angle PQE = 90^\circ$ or $\angle PQF = 90^\circ$.

- 5 Let O be the circumcenter of an acute triangle ABC . Line OA intersects the altitudes of ABC through B and C at P and Q , respectively. The altitudes meet at H . Prove that the circumcenter of triangle PQH lies on a median of triangle ABC .

2.2 Hard

- 1 Let ABC be an acute, scalene triangle with orthocentre H . Let ℓ_a be the line through the reflection of B with respect to CH and the reflection of C with respect to BH . Lines ℓ_b and ℓ_c are defined similarly. Suppose lines ℓ_a , ℓ_b , and ℓ_c determine a triangle $\mathcal{T}$.
Prove that the orthocentre of $\mathcal{T}$, the circumcentre of $\mathcal{T}$, and H are collinear.
- 2 Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcenters of the triangles ADC and EXD , respectively. Prove that the lines BC , EF , and O_1O_2 are concurrent.
- 3 Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC . Let S be the midpoint of the arc CB of Ω containing A . The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L . Denote the circumcircle of triangle BDL by ω . Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.
- 4 Let I be the incenter of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC , CA , and AB at D , E , and F , respectively. The line through D perpendicular to EF meets ω at R . Line AR meets ω again at P . The circumcircles of triangle PCE and PBF meet again at Q .
Prove that lines DI and PQ meet on the line through A perpendicular to AI .
- 5 Let ABC be an acute triangle with circumcircle Γ . Let ℓ be a tangent line to Γ , and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC , CA and AB , respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ .
- 6 Let ABC be an acute triangle. Let M be the midpoint of side BC , and let E and F be the feet of the altitudes from B and C , respectively. Suppose that the common external tangents

to the circumcircles of triangles BME and CMF intersect at a point K , and that K lies on the circumcircle of ABC . Prove that line AK is perpendicular to line BC .